1. Complete parts (a) and (b) below.
a. Write the inverse of $f(x) = 2x - 8$.
b. Do the values for f^{-1} in the provided table fit the equation for f^{-1}?

x	f(x)
−1	−10
0	−8
1	−6
2	−4
3	−2

x	$f^{-1}(x)$
−10	−1
−8	0
−6	1
−4	2
−2	3

a. The inverse of f(x) is $f^{-1}(x) =$ ______.
(Simplify your answer. Use integers or fractions for any numbers in the expression.)

b. Do the values for f^{-1} in the provided table fit the equation for f^{-1}?

- Yes
- No

2. Find the inverse of $g(x) = 6x + 4$.

$g^{-1}(x) =$ ______ (Simplify your answer.)

3. Consider the table giving values for the forward and inverse functions.

$f(-6) = 8$

$f^{-1}(7) = -7$

$f(0) = -7$

$f(7) = 9$

$f^{-1}(-6) = 9$

$f^{-1}(0) = 8$

Find the values of the following.

$f(8) =$ ______

$f^{-1}(8) =$ ______

$f(-7) =$ ______

$f^{-1}(-7) =$ ______

$f(9) =$ ______

$f^{-1}(9) =$ ______

4. The function that models the percent of children taking antidepressants from 2004 to 2009 is $f(x) = -0.084x + 2.93$, where x is the number of years after 2000.

a. Find the inverse of this function. What do the outputs of the inverse function represent?
b. Use the inverse function to find when the percentage is 2.4%.

a. The inverse function is $f^{-1}(x) =$ ______.

What do the outputs of the inverse function represent? Choose the correct statement.

- **A.** The inverse function gives the number of years it will take for the percentage of children taking antidepressants to increase by x.
- **B.** The inverse function gives the percentage of children taking antidepressants when the number of years after 2000 is equal to x.
- **C.** The inverse function gives the number of years after 2000 when the percentage of children taking antidepressants is equal to x.

b. In what year is the percentage of children taking antidepressants equal to 2.4%?

______ (Round to the nearest whole number as needed.)

5. For many species of fish, the weight W is a function of the length x, given by $W = kx^3$, where k is a constant depending on the species. Suppose $k = 0.005$, W is in pounds, and x is in inches, so the weight is $W(x) = 0.005x^3$.

a. Find the inverse function of this function.
b. What does the inverse function give?
c. Use the inverse function to find the length of a fish that weighs 40 pounds.
d. In the context of the application, what are the domain and range of the inverse function?

---

a. Give the inverse function.

$W^{-1}(x) =$ ▭

b. What does the inverse function give? Choose the correct answer below.

- **A.** Given the length, the inverse function calculates the species constant, k.
- **B.** Given the weight, the inverse function calculates the species constant, k.
- **C.** Given the weight, the inverse function calculates the length.
- **D.** Given the length, the inverse function calculates the weight.

c. A fish that weighs 40 pounds is ▭ inches long.

d. Choose the correct domain and range of the inverse function below.

- **A.** The domain is $x < 0$ and the range is $W^{-1}(x) > 0$.
- **B.** The domain is $x > 0$ and the range is $W^{-1}(x) > 0$.
- **C.** The domain is $x > 0$ and the range is $W^{-1}(x) < 0$.
- **D.** The domain is $x < 0$ and the range is $W^{-1}(x) < 0$.